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A systematic method for spatio-temporal phenology estimation of paddy rice using time series Sentinel-1 images

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ABSTRACT

Spatio-temporal phenological information is essential for regional crop monitoring and accurate management. This study aimed to obtain the spatio-temporal distribution of rice phenology using time series Sentinel-1A images. In this study, a systematic method for spatial and temporal phenological estimation was developed based on multi-temporal Sentinel data. This method consisted of: (1) the Spectral Similarity Measures (SSM) approach for rice mapping, and (2) the DTHT-PF approach, which includes a dynamic threshold algorithm (DT), a hierarchical tree (HT), and a particle filter (PF) used for phenology estimation. Thereinto, a dynamic threshold algorithm for transplanting date retrieval and a hierarchical tree for obtaining an initial phenology, which is then run through a PF, are proposed based on the SAR backscatter time series. In brief, we found that the VH backscatter has the potential to provide accurate phenological estimations for paddy rice. Furthermore, rice mapping at a high accuracy level of 92.97%, transplanting date retrieval, and initial phenology estimations lay a good foundation for understanding the spatio-temporal distribution of rice phenology. A determination factor (R^2) and a root mean square error (RMSE) obtained using the DTHT-PF method were much higher (approximately 0.52 in the difference) and lower (approximately 14.67 in the difference), respectively, than those obtained using the PF approach with uniform initialization. R^2 and RMSE values obtained using the real-time DTHT-PF method are slightly different from those obtained using the non-real-time DTHT-PF method. Map analysis revealed that the transplanting dates and the spatial distribution phenology were larger in the west and north of the study area near the South China Sea because of higher temperatures and earlier transplanting. Altogether, our results suggest that the DTHT-PF method and the VH backscatter obtained from the Sentinel-1A images have great potential for furthering our understanding of the spatio-temporal distribution of phenology estimation. This information can greatly contribute to the improvement of precision agriculture, including crop monitoring and yield predictions.

1. Introduction

As one of the most important sources of food for humans, rice is widely planted in Asia and its range has been extending gradually across Africa and Latin America (De Bernardis et al., 2015a). Rice has been the foundation for the subsistence and development of many human cultures. This is because food security directly affects social stability. Obtaining accurate information regarding the phenological stages of rice (Zheng et al., 2016) helps support water, fertilisation, and crop-dusting management decisions, which in turn facilitate more efficient cultivation practices. For instance, panicle fertilizer at the early reproductive stage has greatly impacted rice grain yield and quality (Ling et al., 2007; Zheng et al., 2016). Additionally, studying phenology contributes to accurate crop mapping (Bargiel, 2017; Dong et al., 2015, 2016; Rad et al., 2019; Zhang et al., 2014; Zhong et al., 2016), monitoring of crop lodging, assessment of net primary production (Kaduk and Heimann, 1996; Wang et al., 2017), and provision of time boundary conditions for crop yield models (Jin et al., 2019). Obtaining accurate knowledge on crop phenology is urgently required for agricultural management so as to boost crop yields given the current high global demand for food.

Remote sensing technology has been widely applied in crop

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monitoring, such as crop-type mapping, phenology estimation, and yield predictions, as it can be used to monitor crops on a regional or even global scale. This technology is more advantageous than traditional ground-based monitoring. Synthetic aperture radar (SAR) is well suited to monitoring crop, because its all-weather and all-day capabilities are irreplaceable especially in regions with cloudy weather. In addition, SAR is sensitive to the structural and dielectric properties of the target surface, which has a proven ability to monitor paddy rice (Nelson et al., 2014). Moreover, Sentinel-1 images, which have fine spatio-temporal scales, are utilized for crop monitoring and tracking the dynamics of crop growth (Arias et al., 2020; Chauhan et al., 2020; Fikriyah et al., 2019; Khabbazan et al., 2019; Kussul et al., 2017; Lasko et al., 2018; Mestre-Quereda et al., 2020; Minh et al., 2019; Nguyen et al., 2016; Veloso et al., 2017; Yang et al., 2018). Therefore, we used time series Sentinel-1 images as the major data source for rice phenology estimation in this study.

Currently, special efforts have been made to collect different data sources and explore various approaches in the estimation of paddy rice phenological states. Phenology estimation techniques can be divided into two groups: (1) techniques that collect data only after harvest (cultivation is finished, non-real-time), and (2) techniques that collect data in real-time. In the first group, some smooth curves, which are fitted by wavelet transformations, Fourier transformations, Savitzky-Golay filtering, asymmetric Gaussian models, double logistic models and/or Whittaker filters etc., are applied to phenological stage estimation by detecting the maximum point, minimal point, and/or the inflection point using certain indexes. These indexes are obtained mainly from the Advanced Very High Resolution Radiometer (AVHRR), the Moderate Resolution Imaging Spectroradiometer (MODIS), and the Système Pour L'Observation de la Terre (SPOT)-VEGETATION(VGT). The indexes obtained include the following: the normalized difference vegetation index (NDVI) (Hmimina et al., 2013; Liu et al., 2017; Onojeghuo et al., 2018; Pan et al., 2015; Tornos et al., 2015), the leaf area index (LAI) (Guan et al., 2014; Yang et al., 2017), the Enhanced Vegetation Index (EVI) (Boschetti et al., 2017; Sakamoto et al., 2005, 2006; Zhang et al., 2003), the Normalized Difference Water Index (NDWI) (Tornos et al., 2015), the Terrestrial Chlorophyll Index (MTCI) (Atkinson et al., 2012), and the Normalized Difference Yellow Index (NDYI) (Dandrimont et al., 2020). These estimations are only made after rice has been harvested and so are closely linked to yield predictions and terrestrial carbon budgets. In the second group, the Kalman filter (KF), extended Kalman filter (EKF), and the Particle filter (PF) for phenology estimations are online or real-time methods that use NDVI (De Bernardis et al., 2015b, 2016) and/or polarimetric parameters (De Bernardis et al., 2015a; Kucuk et al., 2015, 2016; Mandal et al., 2018; Vicente-Guijalba et al., 2014, 2015), especially the HH/VV ratio derived from TerraSAR-X images (De Bernardis et al., 2016). However, there are two general approaches within this second group: (2a) methods providing estimations based only on the current image, and (2b) methods based on dynamic system theory (combining both the current image and an evolution model). In terms of group 2a, Lopez-Sanchez et al. (2011a, 2012, 2013, 2014, 2015), Mascolo et al. (2016), and Yuzugullu et al. (2015) used a decision tree algorithm based on scattering mechanisms and a Wishart classifier for crop phenology retrieval using polarimetric parameters, such as entropy and alpha from the eigenvector decomposition. However, research by Vicente-Guijalba et al. (2014, 2015) and De Bernardis et al. (2015a, 2015b, 2016) belong to group 2b as they used the KF, EKF, or PF approach to obtain more accurate phenology estimation using time series parameters. In addition, there are methods that provide data on phenological stages within a reduced set of discrete intervals (typically from three to seven main stages) (De Bernardis et al., 2015a; Vicente-Guijalba et al., 2014, 2015; Kucuk et al., 2016), and other methods that can provide phenology data in a continuous scale (De Bernardis et al., 2015b, 2016). Moreover, these approaches can identify every phenological stage at any time using a priori information from previous images or any other auxiliary information. These real-time estimations are not only closely connected to yield predictions and terrestrial carbon budgets but also can be applied to precision farming management. However, key dates tend to be estimated in many studies while continuous phenology estimations are rarely performed. Therefore, for continuous phenology estimation, the Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie (BBCH) scale (Lancashire et al., 1991), which describes crop evolution using a continuous numerical scale from 00 to 99, is used in this study. Moreover, the PF can provide a more reliable estimation than the EKF in real-time estimations of crop phenological states (De Bernardis et al., 2015a). Hence, the PF approach was applied to phenology estimation in this study.

The PF approach cannot provide an accurate phenology estimation for locations where transplanting dates differ greatly over a large geographical area e.g. in Taishan County, China. Different transplanting dates result in different initial phenologies which will be used for the PF approach at different locations within such variable study areas. Therefore, we propose a dynamic threshold algorithm (DT) for transplanting date retrieval and a hierarchical tree (HT) for the initial BBCH prior to initialization of the PF approach. This algorithm is called the DTHT-PF in short, which can be used for accurate real-time or non-real-time phenology estimations. In effect, the main novelty of our study lies in proposing a dynamic threshold algorithm and using Sentinel-1 data without a priori knowledge of the transplanting date. The transplanting date is the basis of the accurate real-time estimation of crop phenology in a large area. The specific objectives of this study are as follows: (i) to obtain and use accurate rice mapping data as base data for estimating the spatio-temporal distribution of phenology, (ii) to retrieve the transplanting date and set the initial BBCH before initialization of the PF approach in analysing the spatial distribution, (iii) to compare the phenology estimation with the uniform initialization for the PF approach, the real-time DTHT-PF method, and the non-real-time DTHT-PF method, and (iv) to map transplanting dates and obtain the spatio-temporal distribution of phenology estimations.

2. Study area and datasets

2.1. Study area

The study area (21°55′ N – 22°12′ N, 112°43′ E – 112°56′ E) is situated in Taishan County in Guangdong province, China, on the northern shore of the South China Sea as shown in Fig. 1. This area is surrounded by mountains on three sides, while one side faces the South China Sea. It has a humid subtropical climate with abundant rainfall and sunshine. The average annual temperature, relative humidity, and precipitation (from years 1981 to 2010) are 22.3 °C, 79 and 2002.6 mm, respectively. The climate and geographical conditions are conducive to rice cultivation. In the study area, rice is cultivated twice a year and is referred to as either early season rice or late season rice.

2.2. Datasets

The satellite and ancillary data used in this study consisted of: (a) Sentinel-1A IW Level 1 (L1) Ground Range Detected (GRD) products with a 12-day repeat cycle of the C-band radar satellite, (b) Sentinel-2A/B Multi Spectral Instrument (MSI) L2A products with a 5-day repeat cycle of the multispectral satellite, and (c) phenological information obtained from field-plot data. The Sentinel data is freely provided by the European Space Agency (ESA). Table 1 conveys the attributes of these datasets. Twenty-seven scenes of Sentinel-1A GRD products were created and available for use for the following dates: January 1, 2019 to November 21, 2019, as shown in Fig. 2. The acquisition dates for the Sentinel-2A/B MSIL2A products were August 8, September 22, and October 12, 2019 with almost cloud-free conditions in the area of interest.

One of the most important tasks is the ground campaign where

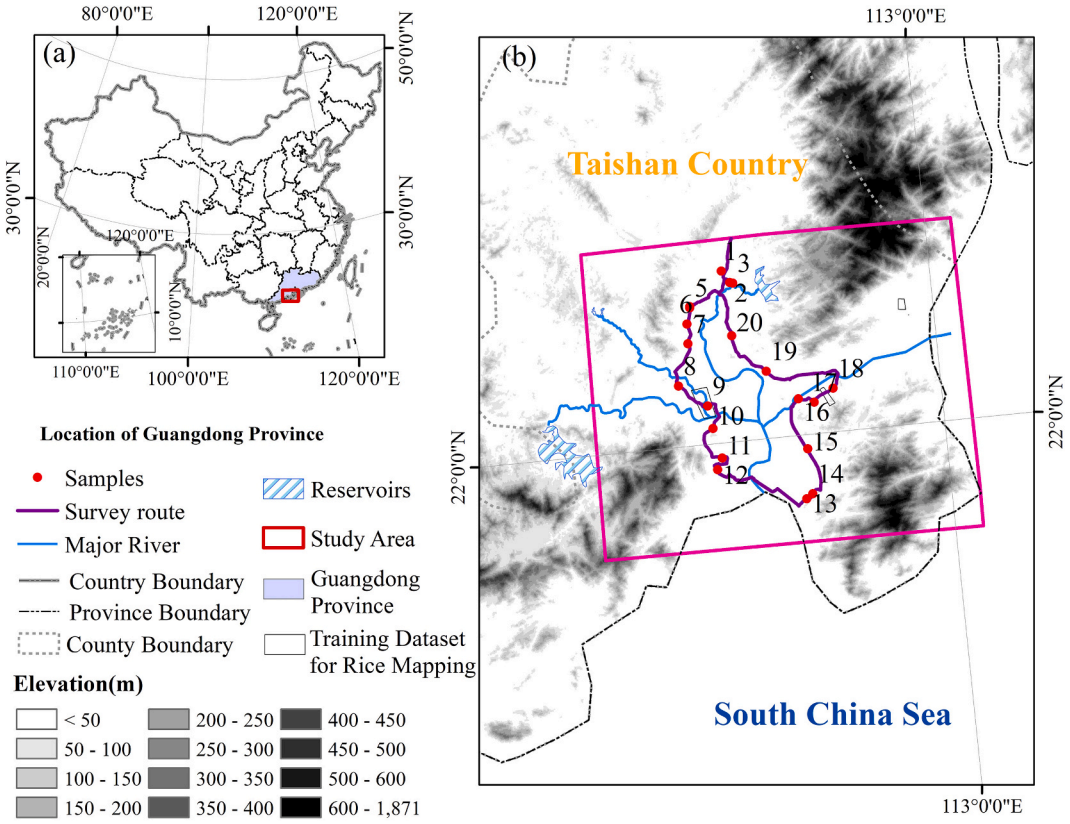


Fig. 1. The location of the study area in Taishan County, Guangdong Province, China. The background in (b) is the 30 m Digital Elevation Model (DEM). Red points are the locations of each field survey. The purple line represents the route of the ground campaigns. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Attributes of Sentinel-1A and sentinel-2A/B images.

Parameters	Sentinel-1A	Sentinel-2A&B		
Product type	GRD		Resolution (m)	Wavelength (nm)
Mode	IW	Band1	60	443.9 ± 45
Polarization	VV, VH	Band2	10	496.6 ± 98
Incidence angle	30.8°-46.2°	Band3	10	560 ± 46
Band	C	Band4	10	664.5 ± 39
Centre frequency	5.4 GHz	Band5	20	703.9 ± 20
Look direction	Right	Band6	20	740.2 ± 18
		Band7	20	702.5 ± 20
Pass direction	Ascending	Band8	10	835.1 ± 145
		Band8a	20	864.8 ± 33
Range/Azimuth looks	5/1	Band9	60	945 ± 27
		Band10	60	1373.5 ± 76
Resolution	10 m	Band11	20	1613.7 ± 143
		Band12	20	2202.4 ± 242

phenological information is collected using field data. Field locations were identified during field surveys using GPS and Google Earth. The specific campaign route and precise site locations are shown in Fig. 1. Data collection was conducted on dates that corresponded to the dates of the Sentinel-1A acquisition as shown in Fig. 2. Nine detailed ground campaigns of approximately 12 days in length were completed during the three mouths. A total of 28,332 pixels were obtained representing 20 paddy fields within the nine ground campaigns. A total of 180 pixels, where each pixel was extracted randomly from each field from nine images, were used as training samples for rice mapping and phenology estimations. The remaining pixels (28152) were used as test samples to validate the model for phenology estimations. The irrigated rice in each field was of the same species and had similar phenological stages. In addition, three regions containing rice and non-rice fields (Fig. 1), identified using Google Earth and field surveys, were used for the verification of rice mapping.

The phenological information of the rice (Fig. 3) recorded during the ground campaigns, corresponded to the scale defined by Meier (2001), which specifically describes the standard BBCH scale with a range of 0–99 (Table 2). The phenology of each single field at each date corresponded to a dominant BBCH since the area of this phenological stage accounted for the largest proportion of the total field area.

3. Methodology

The methodology mainly consisted of five parts: (i) the pre-processing of satellite images, (ii) rice mapping, (iii) transplanting date retrieval, (iv) initial BBCH setting, and (v) phenology estimation. The basic requirement regarding input data for this methodology were time-series images obtained at sampling intervals of 12 days. Transplanting date retrieval and initial BBCH settings were based on a square window of 10 × 10 pixels and were used as the input data for the PF phenology estimation analysis.

3.1. Pre-processing of Sentinel-1&2 images

Sentinel-1A L1 GRD products, i.e. without phase information, were detected, multi-looked, and projected to a ground range using an Earth ellipsoid model by the ESA. Pre-processing is indispensable for Sentinel-1A GRD products, and the pre-processing chain comprises 5 steps. (1) The application of orbit correction provides accurate satellite position and velocity information; (2) calibration correction provides imagery in which the pixel values of the SAR images truly represent the radar

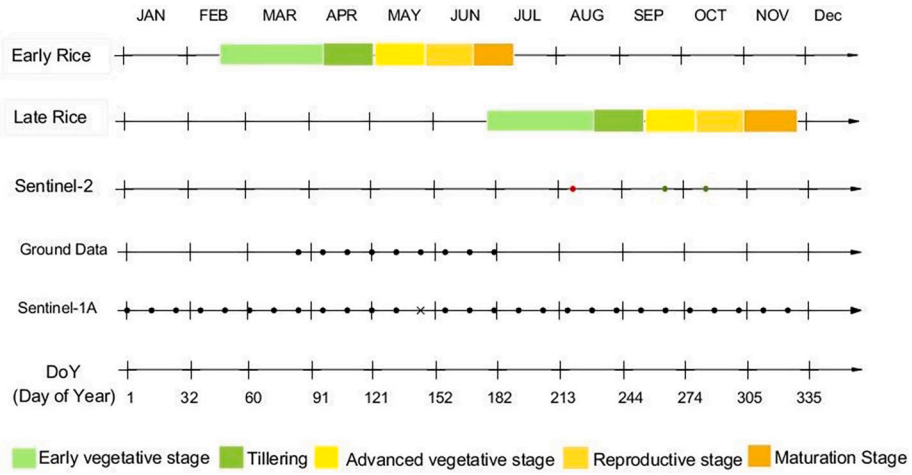


Fig. 2. Rice cropping calendar and schedule of acquisitions organized as Sentinel images and ground campaigns. Black circles denote acquired Sentinel-1A images, and crosses denote a lack of images. The red circle denotes a Sentinel-2A image, and green circles denote Sentinel-2B images. There are no Sentinel-1A images with the acquisition date of May 25, 2019. The backscattering coefficients of rice for May 25, 2019 were calculated based on the mean value of backscattering coefficients for May 13 and June 5, 2019. The early vegetative stage mainly comprises germination and leaf development; the advanced vegetative stage mainly comprises stem elongation and booting; the reproductive stage mainly comprises heading and flowering; and the maturation stage mainly comprises ripening and senescence (Lopez-Sanchez, J.M., et al., 2014). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

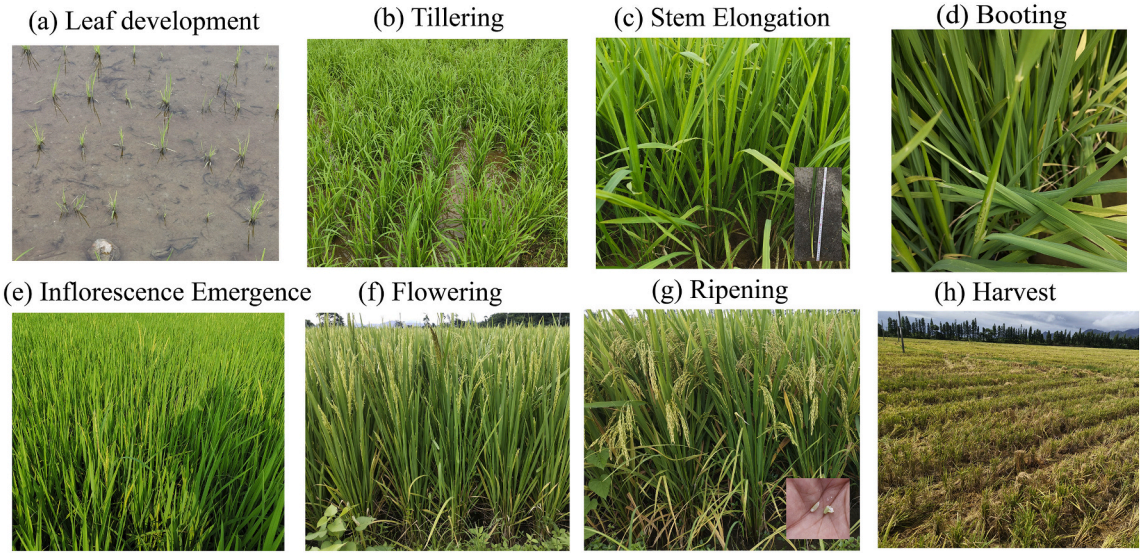


Fig. 3. The phenological growth stages recorded during the ground campaigns. The BBCH of images are: (a) 13 with three unfolded leaves, (b) 24 with four detectable tillers, (c) 35 with elongate internodes, (d) 47 with opening flag leaf sheaths, (e) 55 with middle of panicle emergences, (f) 65 with more anthers, (g) 86 with green grains, and (h) 100 with harvested rice. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
BBCH of growth stages for rice except germination.

BBCH Stage	Description	BBCH Stage	Description
0×	Germination	5×	Inflorescence emergence, heading
1×	Leaf development*	6×	Flowering, anthesis
2×	Tillering	7×	Development of fruit
3×	Stem elongation	8×	Ripening
4×	Booting	9×	Senescence

- The sowing–transplanting period (00–09), Vegetative stage (10–49), Reproductive stage (50–79), Ripening stage (80–99).
-* Tillering may occur earlier than stage 17; in this case continue with stages 21 (Meier, 2001). And secondary code 9 represents completion of the stage (Lancashire et al., 1991).

backscatter of the reflecting surface; (3) speckle filtering (Refined Lee, 7*7 sliding window) reduces speckle noise and improves the quality of the image; (4) range doppler terrain correction with the Shuttle Radar Topography Mission (SRTM) digital elevation model makes the geometric representation of the image as close possible to the real world; (5) backscatter coefficients are converted from linear to dB scale. Sentinel-2A/B L2A products are ortho-images in the UTM/WGS84 projection and the values of pixels represent Bottom-Of-Atmosphere reflectances in cartographic geometry. In our study, band 4 (Red) and 8 (NIR) of Sentinel-2 images were used to calculate NDVI, and the NDVIs of all the three date-based Sentinel-2 images were used for rice mapping.

3.2. Algorithm of rice mapping

An algorithm for rice mapping is presented in the literature (Yang et al., 2018). In this study, Spectral Similarity Measures (SSM) was the key method used for rice mapping.

3.2.1. Spectral similarity measures

SSM consists of Euclidean Distance Similarity (EDS) (shown in Eqs. (1) and (2)) and Spectral Correlation Similarity (SCS) (shown in Eq. (3)). The Spectral Similarity Value (SSV) is a combined measure of EDS and SCS, as shown in Eq. (4). Specific descriptions of SSM are presented in

the literature (Homayouni and Roux, 2007). The equations used to calculate EDS are as follows:

$$Ed_{orig} = \sqrt{\sum_{i=1}^{n} (t_i - p_i)^2} \quad (1)$$

and

$$Ed = \left(Ed_{orig} - m\right)/(M - m) \quad (2)$$

where p and t are defined as a pixel vector of the test dataset and a pixel vector of the training dataset, respectively; the n is the number of Sentinel-1A images; the VH backscattering coefficient is regarded as the pixel value; m and M are the minimum and maximum of Ed_{orig} values; and Ed is a normalized value with a range of 0 to 1. A set of time-series Sentinel-1A L1 images are regarded as a cube and a pixel is regarded as a vector. The equations used to calculate SCS and SSV are as follows:

$$\rho = \frac{1}{n\text{-}1}\left(\frac{\sum_{i=1}^{n}(t_i - \mu_t)(p_i - \mu_p)}{\sigma_t \sigma_p}\right) \quad (3)$$

and

$$SSV = \sqrt{Ed^2 + (1\text{-}\rho)^2} \quad (4)$$

where μ and σ are the mean and standard deviation of two vectors (training dataset and test dataset). If the SSV of a pixel is less than 0.45, this pixel is regarded as rice.

3.2.2. The processing of NDVI

Firstly, three NDVI images from the Sentinel-2 data were used for unsupervised classification using ISODATA Clustering with eight classes. Then water, buildings, and forest were masked in each image and the image (following this masking procedure) was used for rice mapping (Fig. 4).

Secondly, ISODATA Clustering was used for the segmentation of the previously masked NDVI images, rather than for classification of land cover. After using ISODATA Clustering, the segmented image was converted to numerous polygons and each polygon was regarded as a single field or a natural parcel. Finally, the land cover of a field was regarded as rice when the area of rice in this field, following the SSM approach (see Section 3.2.1), accounted for the largest proportion.

3.3. A dynamic threshold algorithm for transplanting date retrieval

The dynamic threshold real-time algorithm was performed using different length time periods (Fig. 4). Transplanting date retrieval was continuously updated as the number of Sentinel-1A images increased. A vector (σ^o) was extracted from a pixel of a set of time-series Sentinel-1A images based on rice mapping (Section 3.2). Two vectors *(t, Diff)* were then generated from the first vector (σ^o). One of these vectors (t) represented the acquisition dates (the day of year, DoY), which were an arithmetic sequence with a tolerance of 12. The other vector *(Diff(t))* represented a set of the difference between σ^o_{t+12} and σ^o_t. The vector (σ^o) represented the backscattering coefficients of rice, which corresponded to the acquisition dates. Four steps were employed to retrieve the transplant date:

(1) the retrieval of dates that corresponded to backscattering coefficients (σ^o_t) that were less than −18 dB.
(2) the retrieval of dates *{j}*) whose *Diff(j)* was more than *Thr_diff* and whose *Diff(j + 12)* was more than 0 dB. *Thr_diff* that is greater than 0.5 dB represent an unstable threshold. The number of these dates was ≤3. The way to determine this threshold is shown in Fig. 4.
(3) the retrieval of dates (*{ts, ts∈{i}, and ts∈{j}}*) that satisfy the first two steps at the same time.
(4) the verification of those dates (*{ts}*) which satisfy the below conditions (i – iv), followed by the retrieval of the post-transplanting date (PTD, *pt*). If the first date (ts_1) satisfies the below conditions, the loop finishes, otherwise the loop continues. In our study, *pt*-12 was defined as the transplanting date. The conditions used are as follows:
   i. $\sigma^o_{ts-12} < -17$ dB. σ^o_{ts-12} is the backscattering coefficient at the date of DoY *ts-12*.
   ii. σ^o_{tx} ($tx \in [ts + 12, t_{coeff_max}]$) is less than −13 dB and more than −21 dB. t_{coeff_max} is the date corresponding to σ^o_{max}. σ^o_{max} is the maximum backscattering coefficient from DOY *pt* to the latest date.
   iii. *decl_nums* is not more than 3. *decl_nums* is the number of Diff (t) that is less than 0. In addition, t ranged from *pt* to t_{coeff_max}.
   iv. *Diff(tx)* ≥ −2 dB.

The transplanting date needs to be identified indirectly using PTD retrieval (see Section 5.2). Fig. 5 shows the temporal signature for selected representative pixels to visualize distinct features of PTD (DoY *pt*) during the growth stages. Most $\sigma_{VH}{}^o$ from the first (DoY *pt*) and second (DoY *pt + 12*) acquisition date after the transplanting date were less than −17 dB, as shown in Fig. 5. In addition, the difference between the VH backscatters from the first and second acquisition dates after transplanting (*Diff(pt)*), which is the red line in Fig. 5, is the largest of all growth stages due to the presence of more water at the leaf development stage. The difference between the second and third acquisition dates (*Diff(pt + 12)*) was similar. Since the *Diff(pt)* and *Diff(pt + 12)* vary in spatial distribution because of the different heights and densities of paddy rice plants and the different water depths in a paddy field, there is an instability threshold. For instance, the *Diff(pt)* in Fig. 5e is larger than in Fig. 5f. In addition, it is possible that the difference between σ^o_{t+12} and σ^o_t during the reproductive and ripening stages is also within this threshold. Therefore, it is quite difficult to retrieve a more precise PTD. To solve this problem, a dynamic threshold algorithm was developed, and the dynamic threshold was set to be ≥0.5, as shown in Fig. 5f. Time series radar backscattering of grass fields can be similar to rice at the vegetative stage, but it is different during the reproductive and ripening stages. Therefore, to obtain refined transplanting dates, some rules (step 4) at the reproductive and ripening stage are proposed for dealing with outer errors. As shown in Fig. 5a and i, σ^o_{pt-12} should be less than −17 dB; the maximum of σ^o (σ^o_{max}) during growth stages should be less than −13 dB, as shown in Fig. 5c-g; σ^o_{pt+12} should be larger than −21 dB, as shown in Fig. 5f, because the water in paddy fields is almost level with rice leaves; from post-transplant date to harvest, the number of Diff(t) with less than 0 is ≤3, as shown in Fig. 5f; all Diff(tx) are ≥ −2 dB, as shown in Fig. 5c.

3.4. A hierarchical tree for the initial BBCH setting

Following the results of the transplanting date retrieval analysis, three algorithms for the initial BBCH were employed to compare phenology estimation with the uniform initialization used in the PF approach, the real-time DTHT-PF method, and the non-real-time DTHT-PF method. We describe the three algorithms as follows.

(1) The uniform initialization of the BBCH was based on the transplanting date of DoY 85 and an initial BBCH of 16. This was because the BBCH of most paddy fields on DoY 85 was found to be approximately 16 by the ground campaigns.
(2) The initial BBCH setting for the real-time phenology estimation is detailed in Fig. 4. If the sum of *Diff(pt)* and *Diff(pt + 12)* was not less than *4* (the first condition), the initial BBCH was set to *14*. If the sum of *Diff(pt)* and *Diff(pt + 12)* was less than *4* (the second condition), the initial BBCH was set in different ways, as follows.

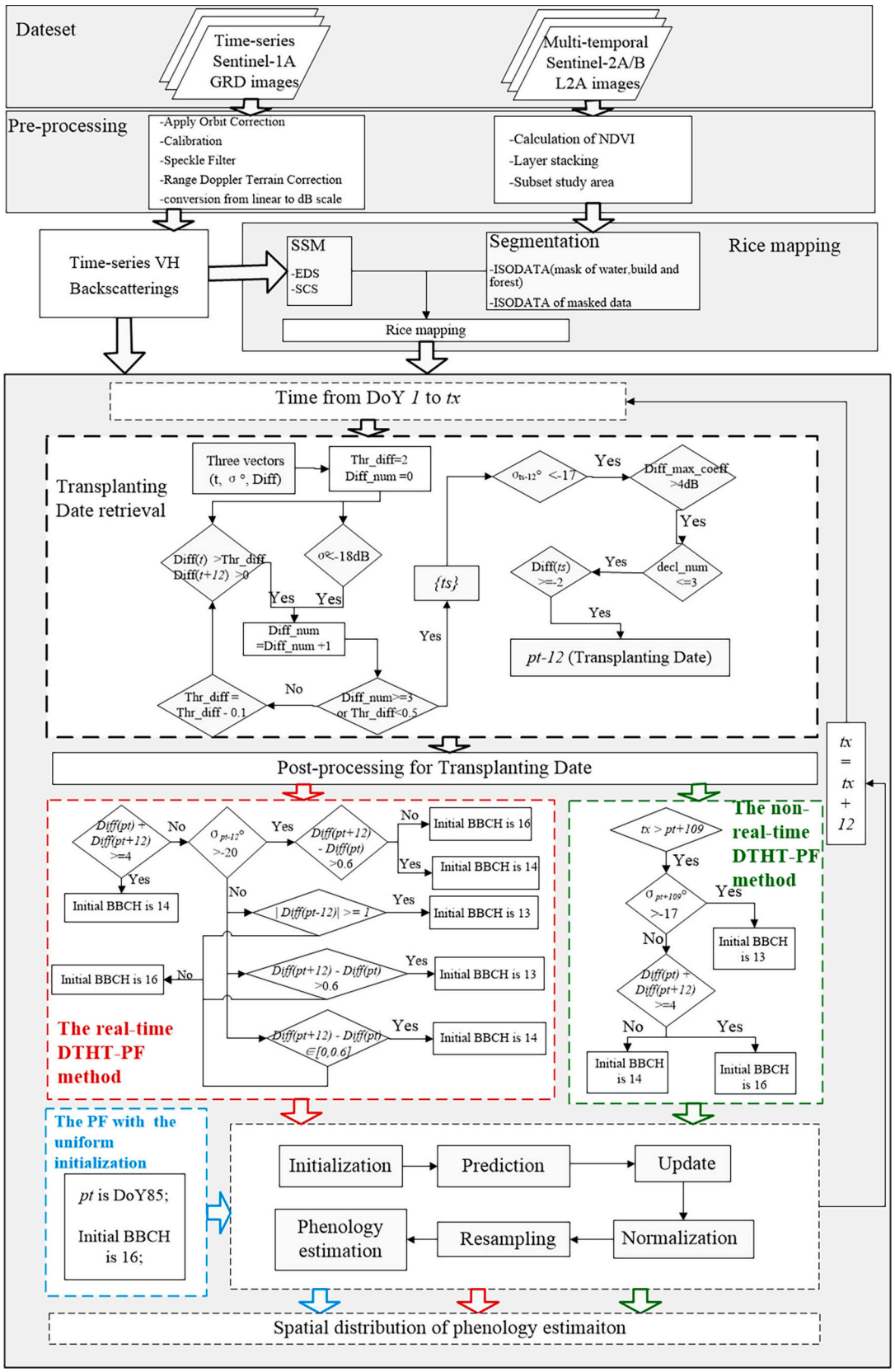


Fig. 4. Overview of the methodology for mapping rice areas, transplanting date retrieval, initial BBCH setting, and phenology estimations. Blue, red, and green arrows represent the PF method using a uniform initialization, the real-time DTHT-PF method, and the non-real-time DTHT-PF method, respectively. Details for each parameter are given in Sections 3.3 and 3.4. The post-processing procedure for transplanting date retrievals is as follows: the transplanting date of one pixel is the same as that of its nearest pixel when a transplanting date of 0 has not been retrieved successfully. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

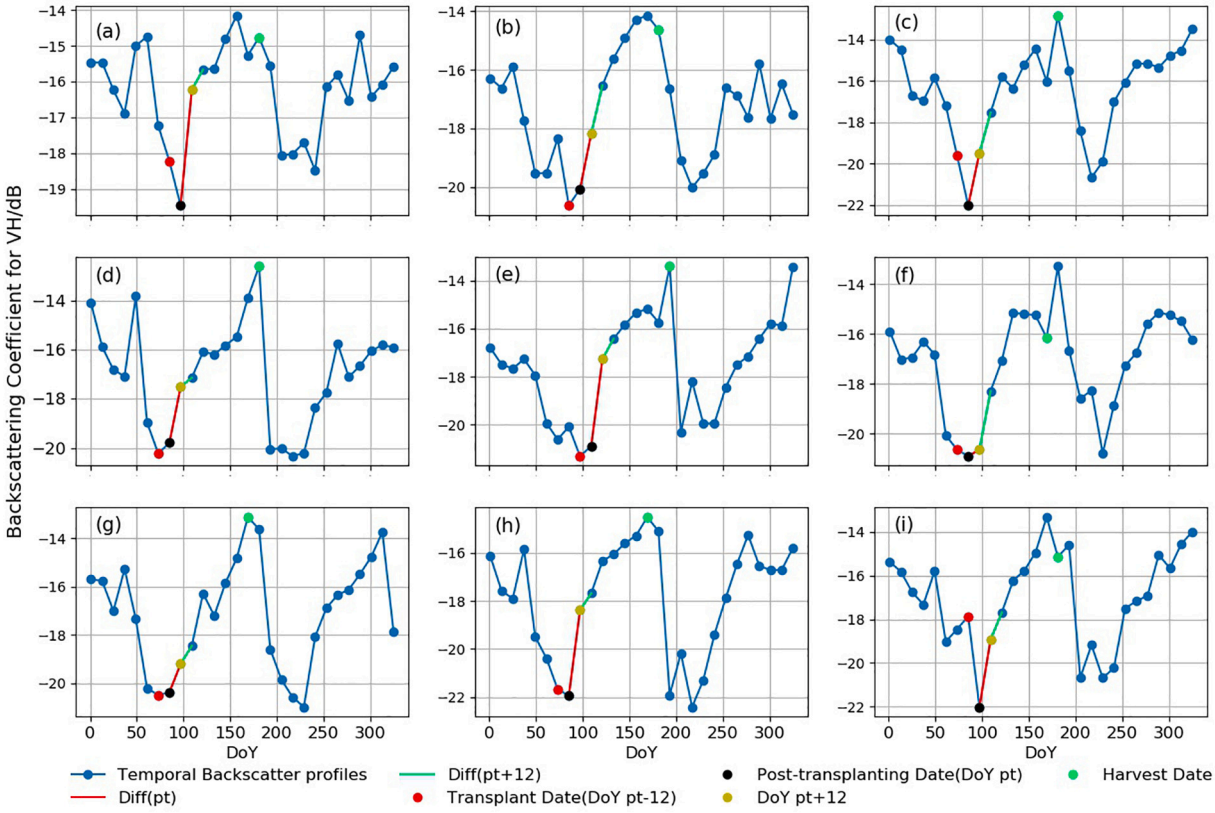


Fig. 5. Temporal backscatter profiles of σ^o_{vh} for rice during all growth stages for different pixels. Red lines represent the difference between σ^o_{pt+12} and σ^o_{pt}, and green lines represent the difference between σ^o_{pt+24} and σ^o_{pt+12}. Red, black, and green points represent transplant date, post-transplant date, and harvest date, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

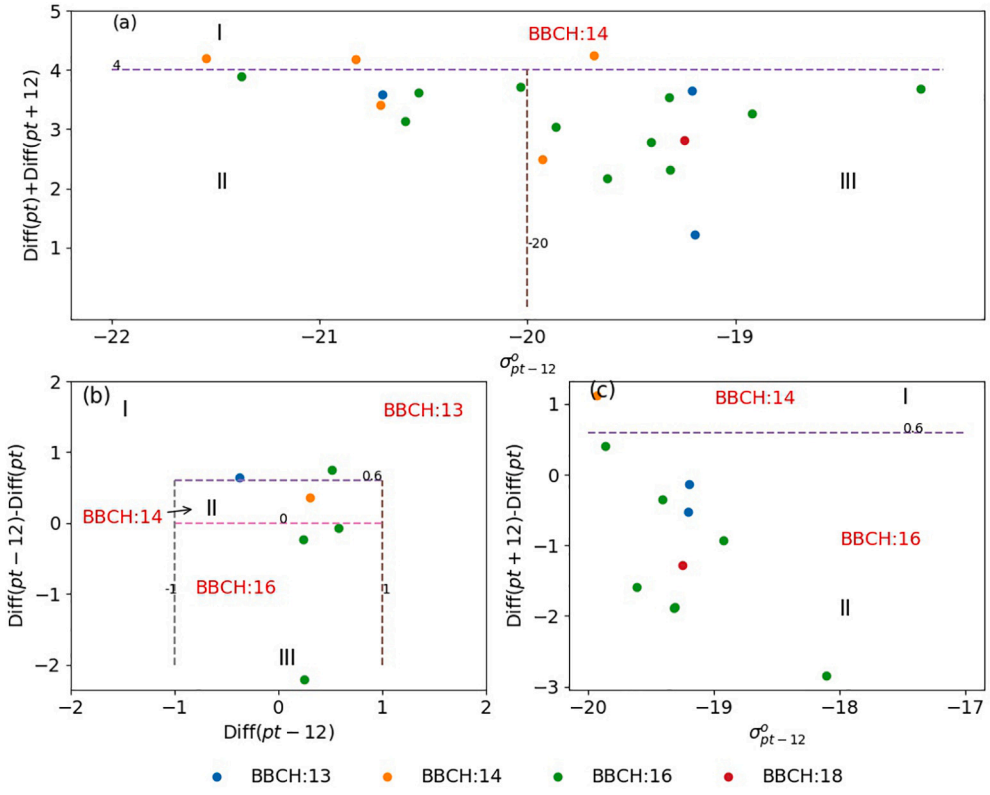


Fig. 6. Different thresholds were extracted for the real-time method by calculating (a) the relationship between σ^o_{pt-12} and the sum of Diff(*pt*) and Diff(*pt* + *12*), (b) the relationship between Diff(*pt* - *12*) and the difference between Diff(*pt* - *12*) and Diff(*pt*), and (c) the relationship between σ^o_{pt-12} and the difference between Diff(*pt* + *12*) and Diff (*pt*). The points in (b) and (c) are the same points shown in zones II and III in (a), respectively. The BBCH in red font were estimated based on the relationship between the backscattering coefficients and the difference between backscattering coefficients. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Under the second condition, there were two sub-conditions: (2a) $\sigma_{pt-12}{}^{o}$ was less than −20 dB, and (2b) $\sigma_{pt-12}{}^{o}$ was not less than *−20* dB. Under the sub-condition 2a, if the absolute value of *Diff (pt - 12)* was not less *1* or the difference between *Diff(pt + 12)* and *Diff(pt)* was larger than *0.6* (2a.i), the initial BBCH was set to *13*. If the difference between *Diff(pt + 12)* and *Diff(pt)* was not less than *0.6* (2a.ii), the initial BBCH was set to *14*. Other values that were beyond the 2a.i and 2a.ii conditions had the initial BBCH set to *16*. Under the sub-condition 2b, if the difference between *Diff (pt + 12)* and *Diff(pt)* was larger than *0.6* (2b.i), the initial BBCH was set to *14*. If the difference between *Diff(pt + 12)* and *Diff(pt)* was not larger than *0.6* (2b.ii), the initial BBCH was set to *16*.

σ^{o}_{pt-12} is less than −20 dB (Fig. 6) when paddy rice is newly transplanted. At this time, radar backscattering is dominated by a double-bounce interaction between flooded soil and quasi-vertical stems (Fig. 7). The sum of *Diff(pt)* and *Diff(pt + 12)* was larger than 4 dB (Fig. 6a and Fig. 8), or the difference between *Diff(pt)* and *Diff(pt + 12)* was larger than 0 dB (Fig. 6b and c). This happened in a situation in which one Sentinel-1A image was obtained when the paddy rice was newly transplanted: paddy rice has an initial fast growth, which is then followed by a slow growth rate during the vegetative phase. The sum of *Diff(pt)* and *Diff(pt + 12)* was smaller than 4 dB (Fig. 6a and Fig. 8), or the difference between *Diff(pt)* and *Diff(pt + 12)* was smaller than 0 (Fig. 6b). This happened when one Sentinel-1 image was obtained when the paddy rice had been transplanted for some time. The reason is the same as above. *Diff(pt - 12)* is smaller than −1 or larger than 1 dB (Fig. 6b). *Diff(pt - 12)* was less than −1 dB because a part of the soil was exposed with low water levels when the rice was newly transplanted (on DoY *pt - 12*) (Fig. 7a), and the soil was covered by water on DoY *pt*. *Diff (pt - 12)* was larger than 1 dB because the soil was covered by water when the rice was newly transplanted (on DoY *pt-12*) (Fig. 7b), and a part of the water was covered by rice on DoY *pt*.

(3) The initial BBCH setting for the non-real-time phenology estimation is conducted only after rice harvest (Fig. 4). If σ^{o}_{pt+108} was not less than *−17 dB* (the first condition), the initial BBCH was set to *13*. If σ^{o}_{pt+108} was less than *−17 dB* (the second condition), the initial BBCH was set in different ways, as follows. Under the second condition there were two sub-conditions: (2a) if the sum of *Diff(pt)* and *Diff(pt + 12)* was not less than 4, the initial BBCH was set to 14; and (2b) if the sum of *Diff(pt)* and *Diff(pt + 12)* was less than 4, the initial BBCH was set to 16. An σ^{o}_{pt+108} of less than −17 dB (Fig. 8) is due to earlier transplanting and an σ^{o}_{pt+108} of more than −17 dB is due to later transplanting. At this time, radar backscattering was dominated by double scattering from the soil and crop residues (Fig. 3 h).

3.5. Algorithm for phenology estimations

Rice phenology is a dynamic process that can be characterized using

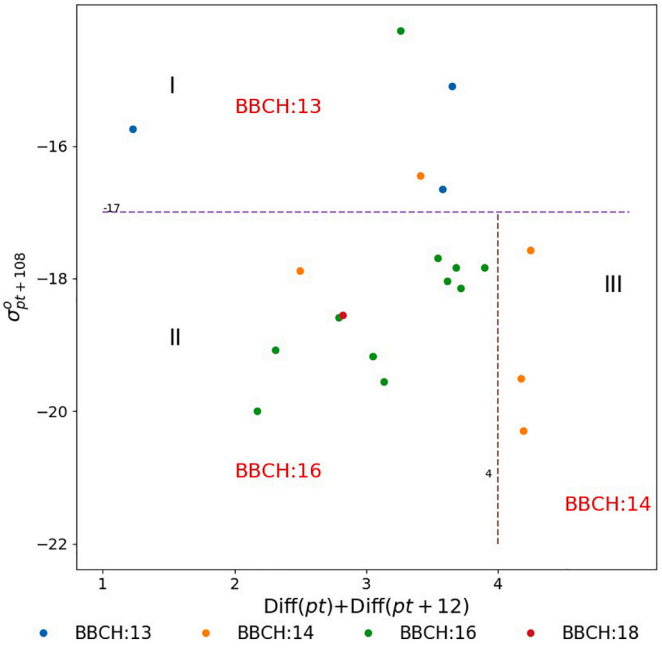


Fig. 8. Different thresholds were extracted for the non-real-time approach based on the relationship between σ^{o}_{pt-108} and the sum of Diff(*pt*) and Diff(*pt + 12*). The BBCH in red font was estimated based on the relationship between the backscattering coefficients and the difference between backscattering coefficients. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

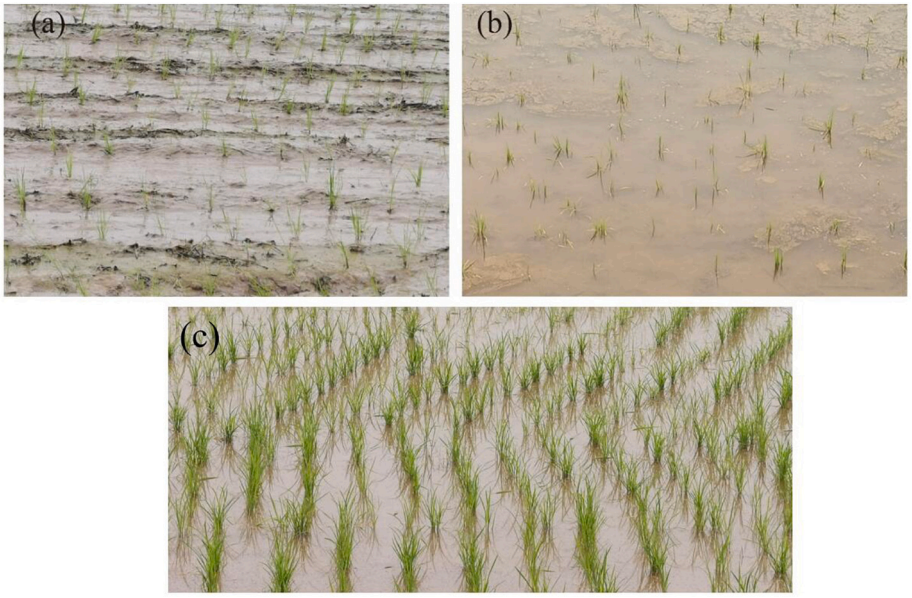


Fig. 7. Photographs taken on the transplanting date (DoY *pt - 12*) at the field 1 (a), field 7 (b) and field 9 (c) during the ground campaigns.

a state space model which is updated with observational data (De Bernardis et al., 2015a, 2015b; Vicente-Guijalba et al., 2014). Therefore, PF, a dynamic method, is used for rice phenology estimations. It consists of a prediction model (Eq. (5)) and an observation model (Eq. (6)). The former characterizes the phenological behaviour of rice fields and the phenological information is updated using the latter (De Bernardis et al., 2015b, 2016). The equations are as follows:

$$x_{k+1} = a\cdot\left(exp(-k\cdot\Delta t)*\left((x_k/a)^{1-d}-1\right)+1\right)^{\frac{L}{1-d}} \quad (5)$$

and

$$y(x_k) = b_0 + b_1\cdot x + b_2\cdot x^2 + b_3\cdot x^3 + b_4\cdot x^4 + b_5\cdot x^5 \quad (6)$$

where x_k is the BBCH at instant k*Δt and $x_{k+1}$is the BBCH at the next instant; *y* is the backscattering coefficient of rice; *a, k, d, b0, b1, b2, b3, b4,* and *b5*, which are the parameters obtained by a fitting procedure to minimize the MSE, are 100.78218, 9.39581, 0.18436, −29.31357, 0.85637, −0.02369, 3.664e-4, −3.05755e-6, and 1.05868e-8, respectively; *t* is the number of days after transplanting; and $\triangle t$, which is the number of days between x_k and x_{k+1}, is 12. The prediction model was fitted by a Sigmoidal Richards (SRichards) model (Richards, 1959) using ground truth data as shown in Fig. 9a. Furthermore, the observation model was better described using a polynomial curve, and its graphical representation is also shown in Fig. 9b. The five main steps of the PF approach are the initialization of particles, the prediction of the sample, the updating of the weight of particles, normalization of the weight of particles, and resampling of these particles. We have provided a brief description in this study. For a more detailed description of this dynamic system and kinds of functions therein, see De Bernardis et al. (2015a, 2015b).

4. Results

4.1. Spectral signatures

Fig. 10 shows the spatial distribution of the NDVI values on August 8 (a), September 22 (b), October 16 (c), accompanied by a RGB image (d). The paddy rice, in our study sites, was newly transplanted on August 8, and the NDVI of paddy rice was found to be mainly between 0 and 0.15. The rice had high NDVI (0.6–0.9) values on September 22 and October 16. Water bodies had the lowest NDVI (< 0) values among these different dates. The NDVI values in urban areas during the three different dates were mainly distributed between 0.1 and 0.4 due to the presence of mixed pixels. In addition, the NDVI values for forest areas were mainly distributed between 0.6 and 0.8 among these different dates. Compared with a single time-point NDVI, the RGB image shows that the three-date NDVI data had a higher separability of land covers and was more suitable for image segmentation.

4.2. Rice mapping

There was little difference between the map of rice created using Sentinel-1A images versus that created using Sentinel-1A and Sentinel-2 L2A images (Fig. 11). The map of rice using Sentinel-1A images was found to have more speckled noise than the map created using Sentinel-1A and Sentinel-2 images (Fig. 12). However, both Fig. 11a and b show that the rice fields located in low altitude regions were mainly located along rivers and near the South China Sea. Moreover, small fragmented patches of rice were concentrated in the north and west of the study area near the headwaters of rivers in mountainous areas, whereas flat and large patches were mainly located in the east and south of the study area near the South China Sea. However, some areas identified as rice (marked using a red circle in Fig. 11) are actually located in the sea due to a measurement error in the SRTM 30 m DEM. Additionally, the total accuracy value of 92.97% (Table 3) observed when using Sentinel-1A and Sentinel-2 images was 2.2% higher than that observed while using only the Sentinel-1A images (Table 4). Therefore, rice mapping using Sentinel-1A and Sentinel-2 images as the base data, is sufficient for accurate phenological stage estimation.

4.3. Maps of transplanting dates

Fig. 13 depicts the last transplanting date retrieval result calculated using the real-time DTHT-PF algorithm. This result is the same as that calculated using the non-real-time DTHT-PF algorithm. The transplanting date estimation of early seasonal rice was in good agreement with the transplanting dates observed using manual interpretation as shown in Fig. 13. In the legend of Fig. 13, the number "-1" indicates that this pixel of paddy rice was not identified, while the number "0" indicates that the transplanting date of this pixel was not retrieved successfully. Thus, the transplanting date of field 1 is −1 due to a failure to differentiate the rice. Moreover, field sizes have effects of misclassifications. As shown in Fig. 13, there are more misclassifications in the fields with small sizes, such as field 1 and field 14. For field 10, transplanting dates were DoY 145 because the single rice was planted later than early seasonal rice. In addition, some pixels with a transplanting date of 0 or > 145 were located along the edges of fields or in the fragmented parcels shown in Fig. 14. In total, the transplanting dates

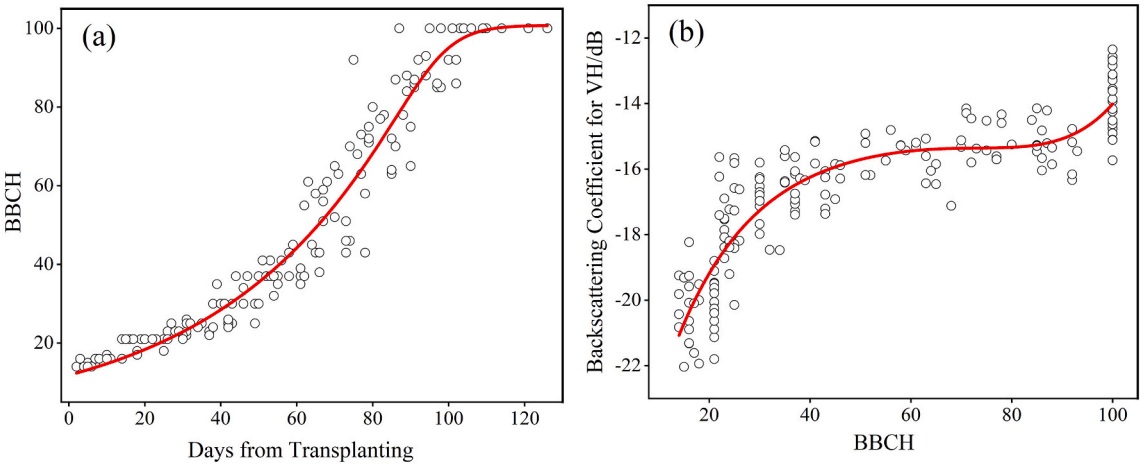


Fig. 9. (a) A prediction model (red line) of phenology in rice crops. (b) An observation equation (red line), which relates phenological states and VH backscatter data. Black circles represent (a) ground truth data or (b) Sentinel-1A VH backscatter data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

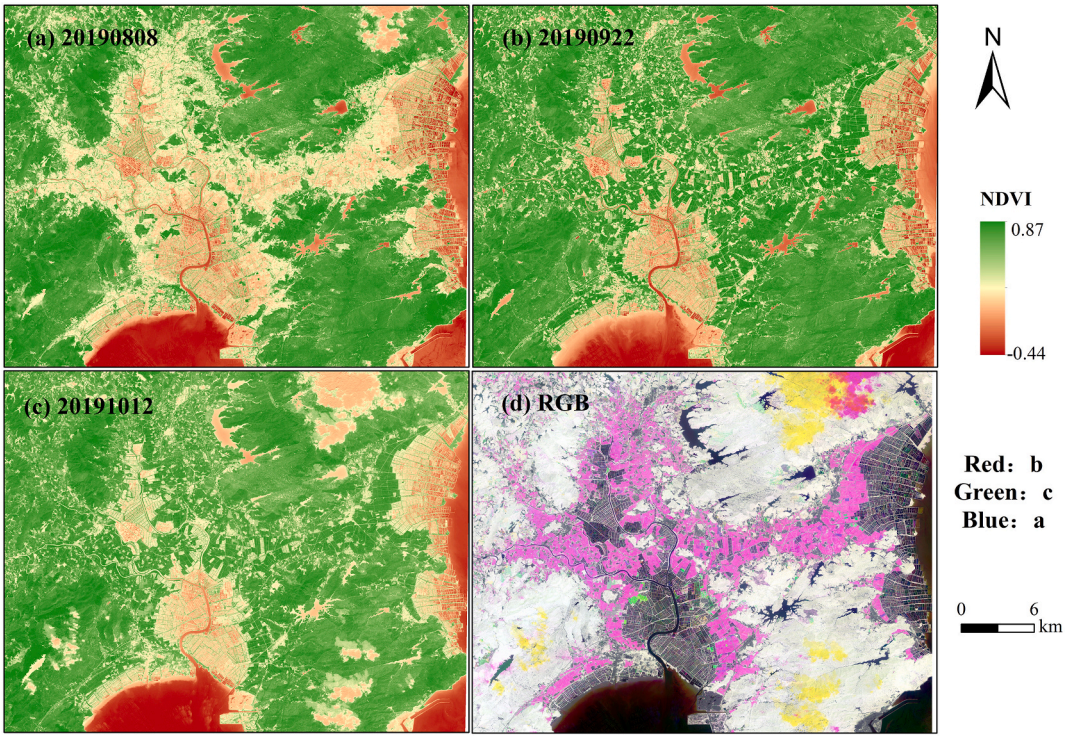


Fig. 10. Spatial distributions of NDVI values derived from Sentinel-2 images taken on August 8 (a), September 22 (b), and October 16 (c), as well as a RGB image (d) (Red: NDVI September 22; Green: NDVI on August 8; and Blue: NDVI on October 16). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

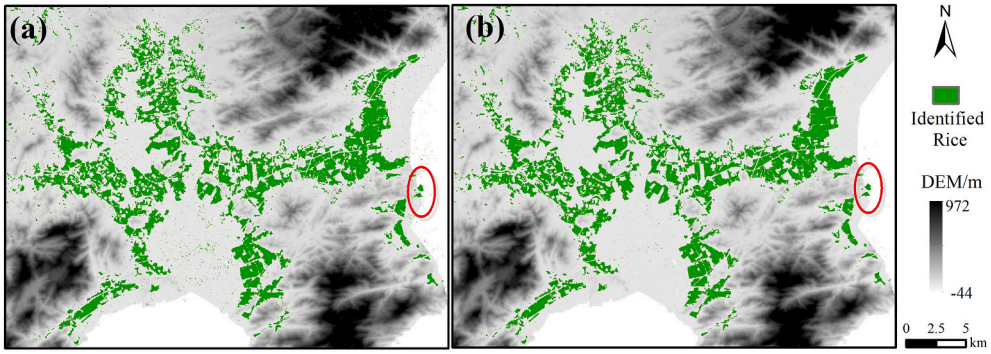


Fig. 11. Maps of rice fields using (a) Sentinel-1A data and (b) Sentinel-1A and Sentinel-2 L2A data. The DEM used SRTM data (http://srtm.csi.cgiar.org/srtmdata/).

of early seasonal rice mainly ranged from 73 to 97 in the study area. In terms of spatial distribution, the transplanting dates in the west and north of the study area were larger than those in the east and south, as shown in Fig. 14. This is because the south and east of the study area near the South China Sea have higher temperatures in early spring. Air temperature is considered a dominant factor driving phenological developments in rice cultivation (Galán Galán et al., 2001; De Bernardis et al., 2016).

Although the dynamic algorithm for the transplanting date retrieval was accessible in real-time, it can only be calculated by obtaining at least two Sentinel-1A images after transplanting, as shown in Fig. 15. With the addition of more images, the accuracy of transplanting date retrieval increased. For instance, the transplanting date on each acquisition date of Sentinel-1A was retrieved for fields 8, 10, and 16 (see Fig. 15). For field 10, there were some pixels recorded as 0 on DoY 145 because the transplanting dates were retrieved on DoY 169. In addition, transplanting dates retrieved using the dynamic threshold algorithm first varied and then gradually stabilised, especially for field 8.

4.4. Spatio–temporal distribution of rice phenology

Fig. 16 shows R^2 and RMSE values for the estimated BBCH for the three aforementioned methods. The R^2 and RMSE values calculated using the DTHT-PF method were approximately 0.52 higher and 14.67 lower, respectively, than those calculated using uniform initialization for the PF approach. The R^2 value calculated using the real-time DTHT-PF method was slightly lower than that calculated using the non-real-time DTHT-PF method. The RMSE value calculated using the real-time DTHT-PF method was slightly higher than that calculated using the non-real-time DTHT-PF method. The points in red circles in Fig. 16 have larger errors because there is a miscalculation regarding the transplanting date (see Fig. 13 for further details). The points in green circles in Fig. 16 have larger errors because rice at field 1 was not correctly identified (see Fig. 13 for further details).

The BBCH estimation can be used for monitoring rice growth. Fig. 17 shows the BBCH estimation for early season rice with nine images made using the real-time DTHT-PF methods during all growth periods. As

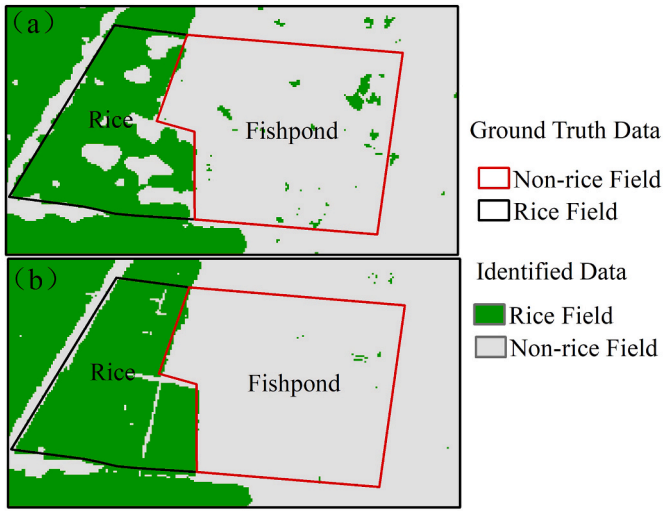


Fig. 12. Comparison between rice discrimination results with the highest total accuracy (a) using Sentinel-1A images and (b) using Sentinel-1A and Sentinel-2 images.

Table 3
Confusion matrix employed for rice mapping using Sentinel-1A images.

	Rice	Non-rice	Total (pixel)	Producer's accuracy (%)
Non-rice	3122	36,179	39,301	92.06
Rice	17,946	2383	20,329	88.28
Total(pixel)	21,068	38,562		
User's accuracy (%)	85.18	93.82		Total: 90.77

Table 4
Confusion matrix employed for rice mapping using Sentinel-1A and Senitnel-2 images.

	Rice	Non-rice	Total (pixel)	Producer's accuracy (%)
Non-rice	2830	36471	39301	92.80
Rice	18968	1361	20329	93.31
Total(pixel)	21798	37832		
User's accuracy (%)	87.02	96.40		Total:92.97

shown in Fig. 17, there was a small change in the BBCH from DoY 85 to 133, while there was a large change in the BBCH from DoY 145 to 169. This is because the vegetative phase takes more time than the reproductive and ripening stages. In terms of spatial distribution, the BBCH in one image was higher than that in the west and south of the study area near the South China Sea, especially on DoY 133, 145, and 169.

4.5. Comparison of three phenology estimation methods

The results of the DTHT-PF methods were superior than those of the PF with uniform initialization, as shown in Fig. 16. Moreover, Figs. 18 and 19 show that the DTHT-PF methods were in excellent agreement in terms of the comparison between the ground truth results and the phenology estimations. Compared with the non-real-time method, results calculated using the real-time method varied in terms of spatial distribution on the same date, especially on DoY 133 and 145, as shown in Fig. 18. The large difference between methods was mainly caused by different transplanting dates and the initial BBCH, especially for fields 10 (Fig. 19). However, the accuracies of phenology estimations for some paddy fields between methods did not change much, such as fields 8 (Fig. 18). This is because the BBCH of these fields on DoY 85 calculated using the prediction model were similar to the initial BBCH calculated using uniform initialization for the PF approach.

In the non-real-time DTHT-PF method, radar backscattering on the transplanting date and on the harvest date was taken into consideration. Yet, only the radar backscattering on the transplanting date was taken into account by the real-time DTHT-PF method. To compensate for this, rules for using the initial BBCH in the real-time method were studied in more detail (See Section 3.4). For instance, if three conditions ($Diff(pt) + Diff(pt + 12) < 4$, $\sigma^o_{pt-12} \geq -20$ dB and $Diff(pt + 12) - Diff(pt) > 0.6$) were met, the initial BBCH was set to *14*. Thus, it was found that the real-time method was slightly more complex than the non-real-time method. In conclusion, transplanting date retrieval and initialization of the BBCH are the most important methods for estimating the spatio-temporal distribution of rice phenology based on the PF approach.

5. Discussion

5.1. Polarization analysis

Fig. 20 shows that, throughout the growing periods, the backscattering coefficients increased first due to increases in the density and height of rice (Lopez-Sanchez et al., 2011b; Toan et al., 1997). Then, they stabilised when there was no significant change in terms of structure. However, this also corresponded to a moment of fast development in terms of phenology (Lopez-Sanchez et al., 2013). After this there was a slight decrease resulting from the gradual drying of plants. At all phenological stages the VV backscatter was larger than the VH backscatter. This is likely due to the stronger attenuation of vertical cylinders (stems) at the VV channel (Lopez-Sanchez et al., 2011b; Toan et al., 1997).

It is notable that the relationship between the ground truth phenology and the backscattering coefficients at the VH channel was stronger than that at the VV channel, as shown in Fig. 20. The most likely reason is that the attenuation of vertical cylinders for the VV signal is more affected by water, topography, and the morphological structure of rice in the canopy than the VH signal. At the vegetative stage, the variance of backscattering coefficients at the VV channel decreased. This may have happened due to changes in the water content and morphological structure of rice. Conversely, at the reproductive stage, the variance remains unchanged because the structure of rice has not significantly changed (Lopez-Sanchez et al., 2013). Fig. 13b shows a larger variance at the vegetative stage than at the reproductive stage at the VV channel. This may be because the VV signal seems to be more affected by changes in standing water conditions (Mansaray et al., 2017; Nguyen et al., 2016). Moreover, the HV/VV ratio and the Radar Vegetation Index (RVI, $4 * \sigma^o_{vh}/(\sigma^o_{vh} + \sigma^o_{vv})$) are used for crop growth monitoring of maize, canola, and wheat (Canisius et al., 2018; Mandal et al., 2020a; Nasirzadehdizaji et al., 2019). However, our results indicate that the HV/VV ratio and the RVI are not suitable for rice phenology estimation (see Fig. 20c and d). In addition, time series NDVI, the HH/VV ratio, and other parameters derived from a set of dual-polarimetric (HH-VV) observables using a PCA transformation have been applied to phenological stage estimation with a PF approach (De Bernardis et al., 2015a, 2015b, 2016). These parameters have been found to have a good relationship with phenology in rice crops, but studies are rare in terms of phenology estimation using only the VH signal. In our study, VH backscatter had a better fit with the phenology results, as shown in Fig. 20a. Therefore, we recommend that the VH backscatter parameter be used for rice phenological stage estimation in future studies.

5.2. Motivation for transplanting date retrieval and initialization of BBCH

The transplanting dates of all the paddy rice fields in the spatial distribution dataset were not the same and this changed further when

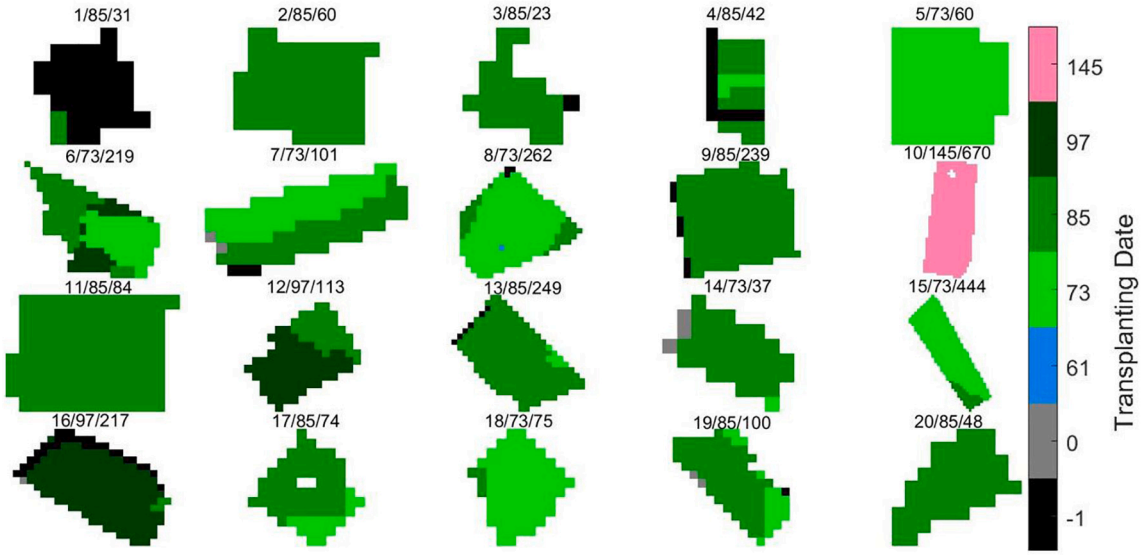


Fig. 13. Transplanting dates for 20 fields. The numbers on the top of every image represent the field code, transplanting date by manual interpretation and the number of pixels in each field. The number "-1" indicates that this pixel of rice was not identified. The number "0" indicates that the transplanting date of this pixel was not retrieved successfully.

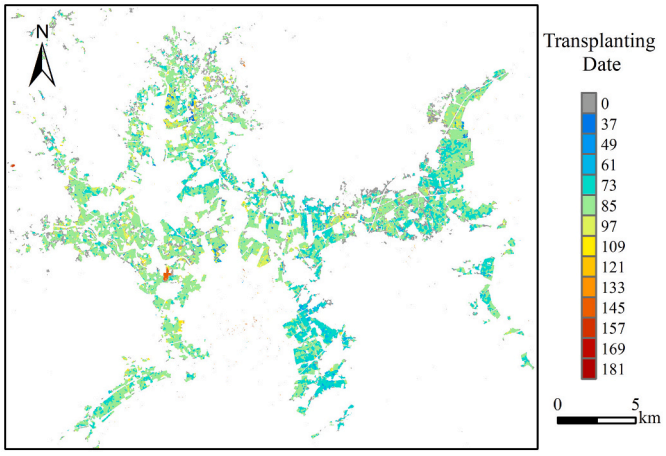


Fig. 14. Spatial distribution of transplanting date estimations for early season rice.

the phenological stage estimation was computed. This was a problem that we needed to overcome. In this study, the first acquisition date of the Sentinel-1A after transplanting was defined as the transplanting date (DoY *pt-12*). It is essential to decrease errors in retrieving the transplanting date. Although the transplanting date was based on a VH backscatter time series, this is not particularly distinguished in its features. The post-transplanting date (PTD), which is the first acquisition date (DoY *pt*) of the Sentinel-1A after the transplanting date, has more obvious features as shown in Fig. 5. Therefore, the transplanting date needs to be identified indirectly using PTD retrieval.

The BBCH values which corresponded to the transplanting date following transplanting date retrieval were not the same in the spatial distribution dataset. Because there are two different situations via the ground campaigns. The first situation was that paddy rice from different transplant dates (the first acquisition date of Sentinel-1A after transplanting) have similar initial BBCH values. The second is that paddy rice from the same transplanting dates have different initial BBCH values. Then another problem exists in how to confirm the initial BBCH on DoY *pt-12*. On the basis of flooding feature, radar backscattering and the difference in radar backscattering between two neighbouring dates were taken into account as indicators of the initial BBCH setting. The initial

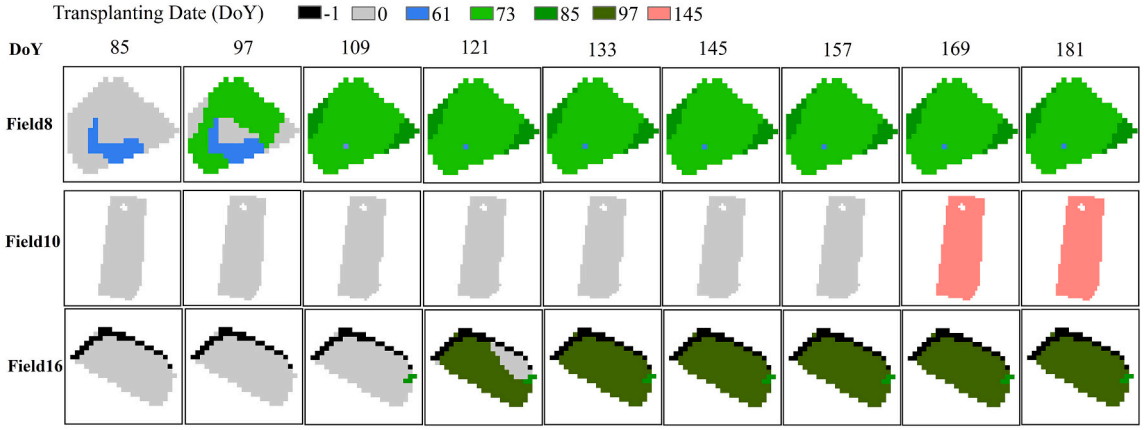


Fig. 15. Transplanting date retrieval for each acquisition date of Sentinel-1A images. *-1* indicates that this pixel of rice was not identified. *0* indicates that the transplanting date of this pixel was not retrieved successfully.

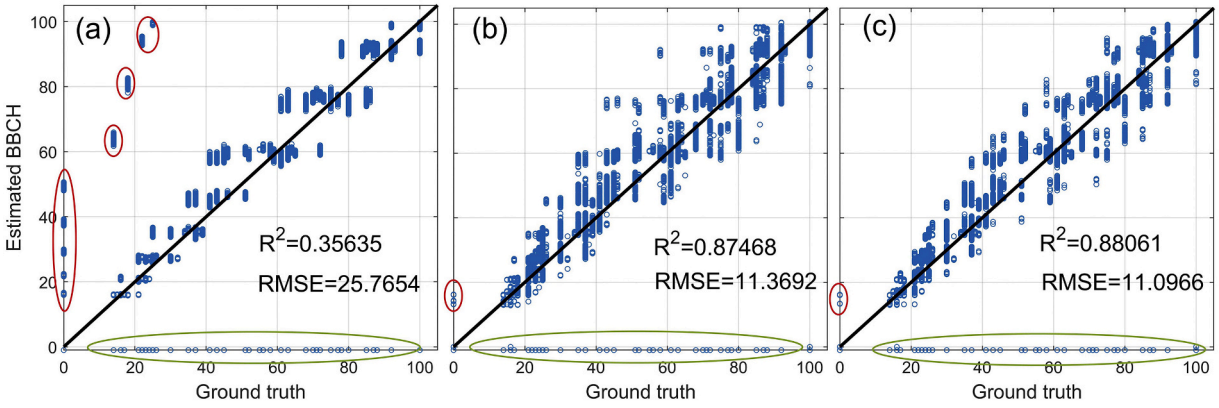


Fig. 16. R^2 and RMSE values of estimated BBCH for three different methods: (a) the PF, (b) the real-time DTHT-PF, and (c) the non-real-time DTHT-PF.

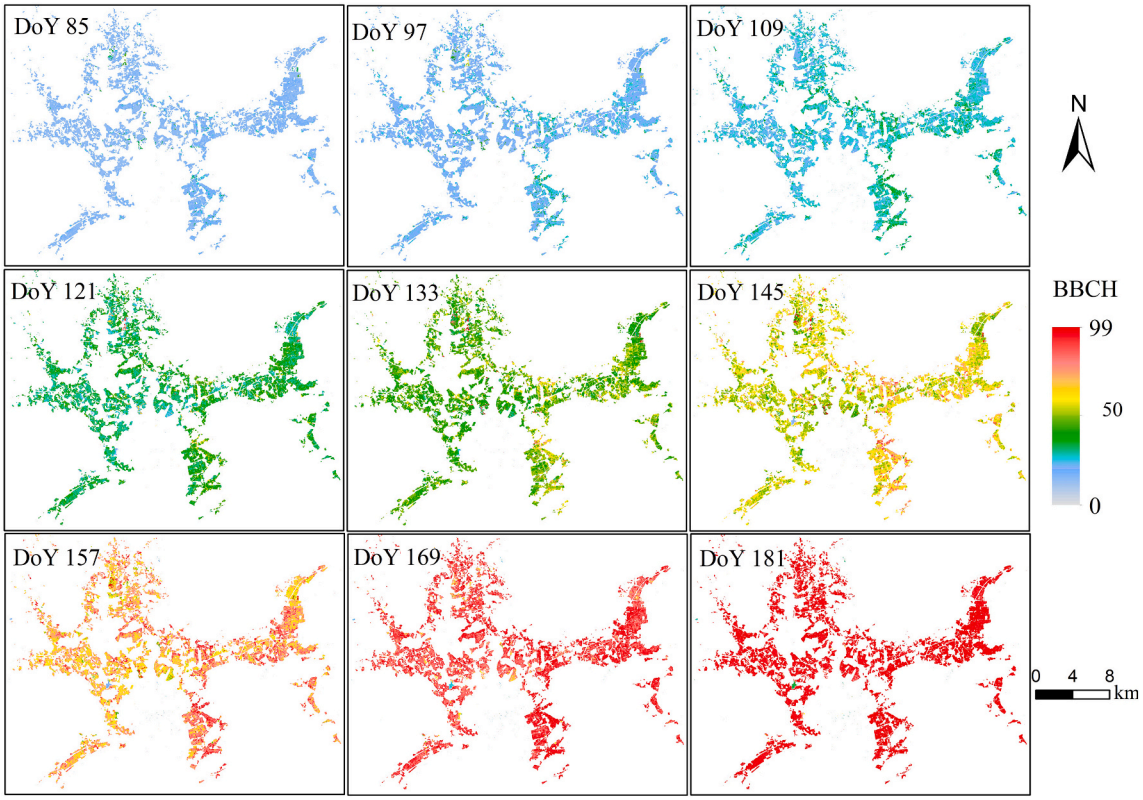


Fig. 17. The spatio–temporal distribution of rice phenology, analysed using the real-time DTHT-PF method.

BBCH were set to 13, 14, and 16 according to newly transplanted paddy rice data or rice that had been transplanted for some time.

Although the initial BBCH was not as important as the transplanting date retrieval, the initial BBCH is indispensable for phenology estimation. There were different initial BBCH values in terms of transplanting date during the ground campaigns (Fig. 7). Moreover, if the initial BBCH was 13 instead of 15 or 16, the results were inaccurate for some fields (Figs. 6 and 8). However, the phenology estimations as a whole were accurate.

5.3. Features for the DTHT-PF method

We found that initialization for PF with the transplanting date is an important parameter used in phenology estimation. In our study, the initial BBCH was obtained using a hierarchical tree based on the transplanting date, which was retrieved by the dynamic threshold algorithm. However, in a study by De Bernardis et al. (2016), the initial BBCH was directly obtained from a uniform distribution of particles between states 0 and 40, and they forgot about the importance of spatial heterogeneity of the transplanting date. Yet the condition, where the transplanting date of rice varies depending on its spatial distribution, was important for our study. Therefore, we propose that the DHTH-PF method be used to improve phenology estimation in spatial distributions. For all growth stages, we conducted real-time phenology estimations when the transplanting dates were retrieved correctly. The results (R^2 = 0.96 and RMSE = 5.82) when using only VH backscattering coefficients were slightly worse than those reported by De Bernardis et al. (2016) (R^2 = 0.97 and RMSE = 3.90) when temperature, NDVI, and SAR data have been taken into consideration. In addition, based on our analysis methods, we were able to report the spatial distribution of rice phenology (Fig. 17), unlike De Bernardis et al. (2016).

For Sentinel-1 images, the incidence angle ranged from 31 to 46

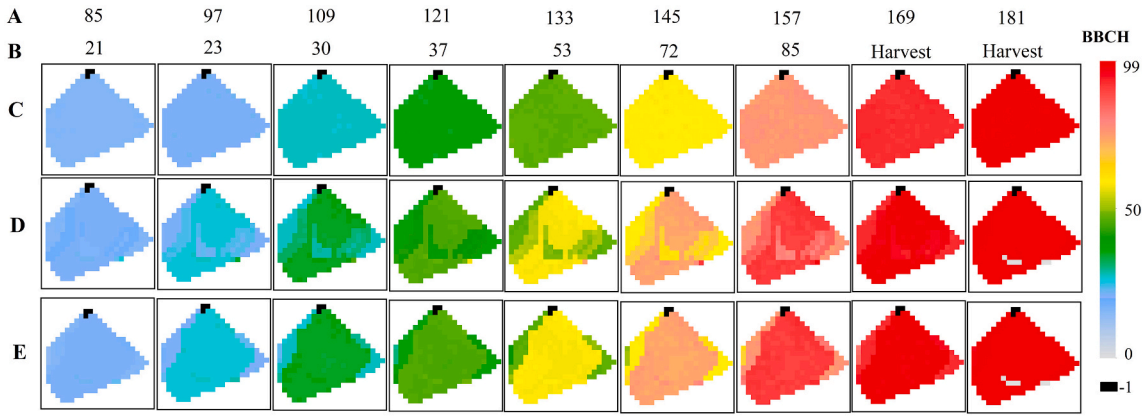


Fig. 18. Phenological estimation for field 8 at the pixel basis using three methods. The first line (A) represents DoY and the second line (B) represents ground truth BBCH. The last three lines (C, D and E) are the results of the PF with uniform initialization, the real-time DTHT-PF algorithm and the non-real-time DTHT-PF, respectively.

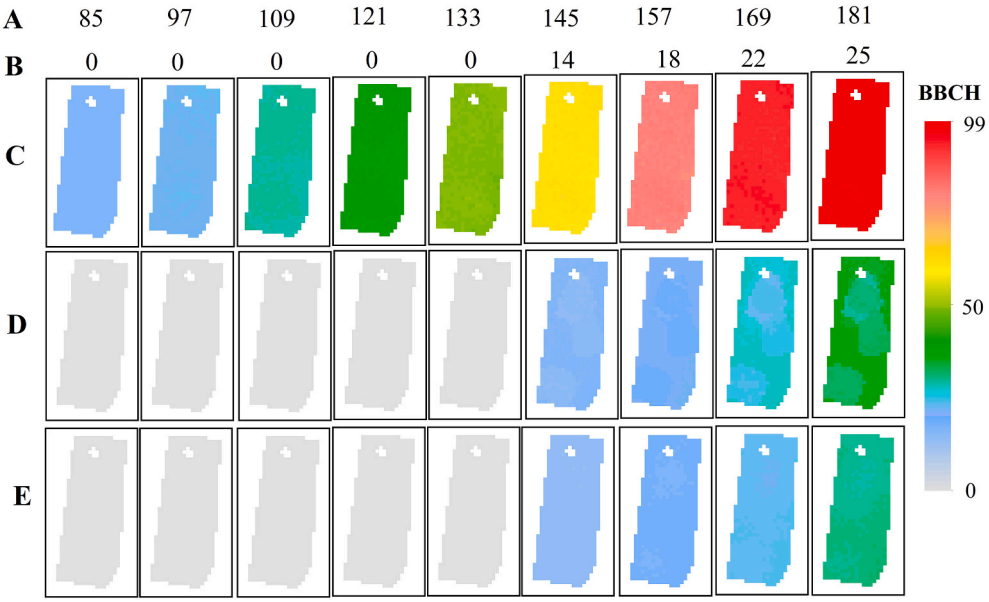


Fig. 19. Phenological estimation for field 10 at the pixel basis using three methods. A, B, C, D and E are the same as above.

degrees. Although backscatter values are affected by the incidence angle especially for the water surface value (Phung et al., 2020), the dynamic threshold algorithm and the hierarchical tree were very robust against transplanting date retrieval and the initial BBCH setting. This was because the volume scattering of paddy rice at the late vegetative, reproductive, and ripening stages changed only slightly across the range (Phung et al., 2020). At the early vegetative stage, σ^o_{vh}, which was less than −17 dB, −18 dB, or − 20 dB in the dynamic threshold algorithm and the hierarchical tree, was not much affected by the incidence angle because the water surface value was between −21 and − 34 dB across the range (Phung et al., 2020). Moreover, *Diff(t)*, which is the difference between σ^o_{t+12} and σ^o_t, changed slightly across the range.

At the late reproductive and ripening stage, σ^o_{vh} only changed slightly due to a lack of structural change. Though backscatter powers remained almost constant, the phenology estimation was in agreement with the ground truth results. We have proven that the prediction model (Eq. (5)) can predict the next phenology (x_{k+1}) using a last known phenology (x_k) and that σ^o_{vh} inputted in Eq. (6) can be used for conducting accurate phenology estimations ($\hat{x}_{k+1}$). However, one question that limits the application of the DTHT-PF method for phenology estimations is that the threshold algorithm and the hierarchical tree were suitable for normal rice growing conditions during all growth periods. If the rice is seriously lodged by wind or is almost flooded after heavy rain, there will lead to some uncertainties in the transplanting date retrieval and the initial BBCH setting. However, backscatter values through interpolation using nearest pixels in temporal or spatial distance instead of the backscattering coefficients of unnormal rice may alleviate the problem for outliers in temporal signature.

6. Conclusion

In this study, a systematic method for estimating the spatio-temporal distribution of rice phenology was developed based on time series Sentinel-1A images. This systematic method included the SSM for rice mapping, a dynamic threshold algorithm for transplanting date retrieval, a hierarchical tree for calculating the initial BBCH, and the PF approach for real-time estimations. The dynamic threshold algorithm and the hierarchical tree are proposed mainly by flooding characteristics in the SAR backscatter time series. Compared with the RVI and the VH/VV ratio, the VH signal has a higher potential to improve paddy rice monitoring. For rice mapping, the total accuracy using Sentinel-1A and Sentinel-2 images was 2.2% higher than when only Sentinel-1A images

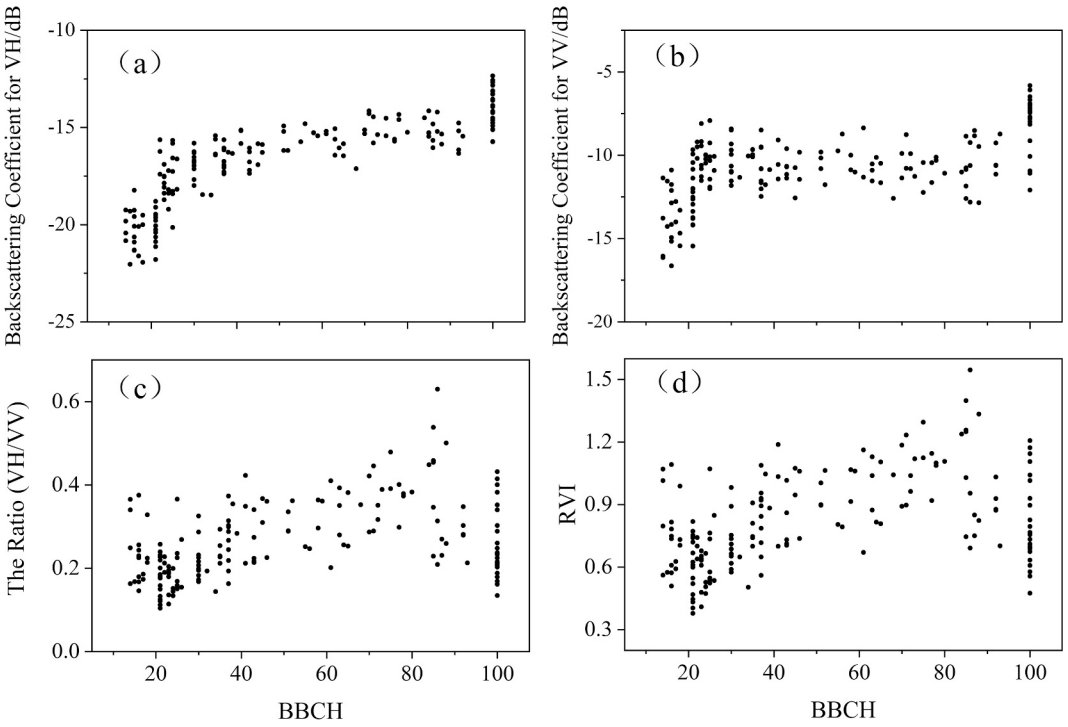


Fig. 20. Relationships between the BBCH and (a) backscattering coefficients for VH, (b) backscattering coefficients for VV, (c) the $\sigma^0_{vh}/\sigma^0_{vv}$ ratio, and (d) the RVI (4 * $\sigma^o_{vh}/(\sigma^o_{vh} + \sigma^o_{vv})$).

were used. Rice mapping with a high accuracy rate of 92.97% for the base data is sufficient to provide accurate phenology estimations. Moreover, the estimated transplanting dates of paddy rice were in excellent agreement with the ground truth results (i.e., by manual interpretation). However, the transplanting date of paddy rice can only be calculated by obtaining at least two Sentinel-1A images after transplanting. If there are more images, transplanting date retrieval will be more accurate and this will lay a good foundation for initialization of the PF approach. Finally, the R^2 (0.88) and RMSE (11.10) values obtained by the DTHT-PF method were approximately 0.52 times higher and 14.67 times lower than those obtained by the PF with uniform initialization method, respectively. The R^2 and RMSE values obtained using the real-time DTHT-PF method were slightly different from those obtained from the non-real-time DTHT-PF method. Analysis of the maps revealed that the transplanting dates and the BBCH, in terms of spatial distribution, were larger in the west and north of study area near the South China Sea because of higher temperatures and earlier transplanting dates. In conclusion, the results suggest that the DTHT-PF method and the VH signal of the Sentinel-1A images have great potential for estimating the spatio-temporal distribution of rice phenology. Our methods, therefore, can greatly contribute to improvements in precision agriculture, crop monitoring, crop yield predictions, and crop health modelling.

Disclosures

The authors declare no conflicts of interest.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Arias, M., et al., 2020. Crop classification based on temporal signatures of Sentinel-1 observations over Navarre Province, Spain. Remote Sens. 12, 278.
Atkinson, P.M., et al., 2012. Inter-comparison of four models for smoothing satellite sensor time-series data to estimate vegetation phenology. Remote Sens. Environ. 123, 400–417.
Bargiel, D., 2017. A new method for crop classification combining time series of radar images and crop phenology information. Remote Sens. Environ. 198, 369–383.
Boschetti, M., et al., 2017. PhenoRice: a method for automatic extraction of spatio-temporal information on rice crops using satellite data time series. Remote Sens. Environ. 194, 347–365.
Canisius, F., et al., 2018. Tracking crop phenological development using multi-temporal polarimetric Radarsat-2 data. Remote Sens. Environ. 210, 508–518.
Chauhan, S., et al., 2020. Estimation of crop angle of inclination for lodged wheat using multi-sensor SAR data. Remote Sens. Environ. 236, 111488.
Dandrimont, R., et al., 2020. Detecting flowering phenology in oil seed rape parcels with Sentinel-1 and -2 time series. Remote Sens. Environ. 239, 111660.
De Bernardis, C., et al., 2015a. Estimation of key dates and stages in Rice crops using dual-polarization SAR time series and a particle filtering approach. IEEE J. Select. Top. Appl. Earth Observ. Remote Sens. 8, 1008–1018.
De Bernardis, C.G., et al., 2015b. Particle filter approach for crop phenological stage estimation using time series of NDVI images. In: International Geoscience and Remote Sensing Symposium, pp. 3385–3388.
De Bernardis, C., et al., 2016. Contribution to real-time estimation of crop phenological states in a dynamical framework based on NDVI time series: data fusion with SAR and temperature. IEEE J. Select. Top. Appl. Earth Observ. Remote Sens. 9, 3512–3523.
Dong, J., et al., 2015. Tracking the dynamics of paddy rice planting area in 1986-2010 through time series Landsat images and phenology-based algorithms. Remote Sens. Environ. 160, 99–113.
Dong, J., et al., 2016. Mapping paddy rice planting area in northeastern Asia with Landsat 8 images, phenology-based algorithm and Google earth engine. Remote Sens. Environ. 185, 142–154.
Fikriyah, V.N., et al., 2019. Discriminating transplanted and direct seeded rice using Sentinel-1 intensity data. Int. J. Appl. Earth Obs. Geoinf. 76, 143–153.
Galán, C., et al., 2001. The role of temperature in the onset of the Olea europaea L. pollen season in southwestern Spain. Int. J. Biometeorol. 45, 8–12.
Guan, K., et al., 2014. Deriving vegetation phenological time and trajectory information over Africa using SEVIRI daily LAI. IEEE Trans. Geosci. Remote Sens. 52, 1113–1130.

Hmimina, G., et al., 2013. Evaluation of the potential of MODIS satellite data to predict vegetation phenology in different biomes: an investigation using ground-based NDVI measurements. Remote Sens. Environ. 132, 145–158.
Homayouni, S., Roux, M., 2007. Hyperspectral Image Analysis for Material Mapping Using Spectral Matching (ISPRS Congress Proceedings).
Jin, Z., et al., 2019. Smallholder maize area and yield mapping at national scales with Google earth engine. Remote Sens. Environ. 228, 115–128.
Kaduk, J., Heimann, M., 1996. A prognostic phenology scheme for global terrestrial carbon cycle models. Clim. Res. 6, 1–19.
Khabbazan, S., et al., 2019. Crop monitoring using Sentinel-1 data: a case study from the Netherlands. Remote Sens. 11, 1887.
Kucuk, C., et al., 2015. CO-POLAR SAR data classification as a tool for real time paddy-rice monitoring. In: International Geoscience and Remote Sensing Symposium, pp. 4141–4144.
Kucuk, C., et al., 2016. Paddy-rice phenology classification based on machine-learning methods using multitemporal co-polar X-band SAR images. IEEE J. Select. Top. Appl. Earth Observ. Remote Sens. 9, 2509–2519.
Kussul, N., et al., 2017. Deep learning classification of land cover and crop types using remote sensing data. IEEE Geosci. Remote Sens. Lett. 14, 778–782.
Lancashire, P.D., et al., 1991. A uniform decimal code for growth stages of crops and weeds. Ann. Appl. Biol. 119, 561–601.
Lasko, K., et al., 2018. Mapping double and single crop paddy rice with sentinel-1A at varying spatial scales and polarizations in Hanoi, Vietnam. IEEE J. Select. Top. Appl. Earth Observ. Remote Sens. 11, 498–512.
Ling, Q., et al., 2007. Precise and quantitative cultivation for high yield in rice. North. Rice 9.
Liu, Y., et al., 2017. Using data from Landsat, MODIS, VIIRS and PhenoCams to monitor the phenology of California oak/grass savanna and open grassland across spatial scales. Agr. For. Meteorol. 237, 311–325.
Lopez-Sanchez, J.M., et al., 2011a. Monitoring and retrieving rice phenology by means of satellite SAR polarimetry at X-band. In: International Geoscience and Remote Sensing Symposium, pp. 2741–2744.
Lopez-Sanchez, J.M., et al., 2011b. First results of rice monitoring practices in Spain by means of time series of TerraSAR-X dual-pol images. IEEE J. Select. Top. Appl. Earth Observ. Remote Sens. 4, 412–422.
Lopez-Sanchez, J.M., et al., 2012. Rice phenology monitoring by means of SAR polarimetry at X-band. IEEE Trans. Geosci. Remote Sens. 50, 2695–2709.
Lopez-Sanchez, J.M., et al., 2013. Estimating phenology of agricultural crops from space. In: Proceedings of the ESA Living Planet Symposium, ESA-SP 722.
Lopez-Sanchez, J.M., et al., 2014. Polarimetric response of rice fields at C-band: analysis and phenology retrieval. IEEE Trans. Geosci. Remote Sens. 52, 2977–2993.
Lopez-Sanchez, J.M., et al., 2015. Retrieving rice phenology with X-band co-polar data: A multi-incidence multi-year experiment. In: International Geoscience and Remote Sensing Symposium, pp. 3977–3980.
Mandal, D., et al., 2018. Sen4Rice: a processing chain for differentiating early and late transplanted rice using time-series Sentinel-1 SAR data with Google earth engine. IEEE Geosci. Remote Sens. Lett. 15, 1947–1951.
Mandal, D., et al., 2020a. Dual polarimetric radar vegetation index for crop growth monitoring using sentinel-1 SAR data. Remote Sens. Environ. 247, 111954.
Mansaray, L.R., et al., 2017. Mapping rice fields in urban Shanghai, Southeast China, using sentinel-1A and Landsat 8 datasets. Remote Sens. 9, 257.
Mascolo, L., et al., 2016. A complete procedure for crop phenology estimation with PolSAR data based on the complex Wishart classifier. IEEE Trans. Geosci. Remote Sens. 54 (11), 6505–6515.
Meier, U., 2001. Growth Stages of Mono-and Dicotyledonous Plants: BBCH Monograph, 2nd ed. Federal Research Centre for Agriculture and Forestry, Berlin/Braunschweig.
Mestre-Quereda, A., et al., 2020. Time-series of Sentinel-1 interferometric coherence and backscatter for crop-type mapping. IEEE J. Select. Top. Appl. Earth Observ. Remote Sens. 13, 4070–4084.
Minh, H.V.T., et al., 2019. Monitoring and mapping of rice cropping pattern in flooding area in the Vietnamese Mekong Delta using sentinel-1A data: a case of an Giang Province. ISPRS Int. J. Geo Inf. 8, 211.
Nasirzadehdizaji, R., et al., 2019. Sensitivity analysis of multi-temporal Sentinel-1 SAR parameters to crop height and canopy coverage. Appl. Sci. 9 (4), 655.
Nelson, A., et al., 2014. Towards an operational SAR-based rice monitoring system in Asia: examples from 13 demonstration sites across Asia in the RIICE project. Remote Sens. 6 (11), 10773–10812.
Nguyen, D.B., et al., 2016. Mapping rice extent and cropping scheme in the Mekong Delta using Sentinel-1A data. Remote Sens. Lett. 7, 1209–1218.
Onojeghuo, A.O., et al., 2018. Rice crop phenology mapping at high spatial and temporal resolution using downscaled MODIS time-series. Gisci. Remote Sens. 55, 659–677.
Pan, Z., et al., 2015. Mapping crop phenology using NDVI time-series derived from HJ-1 A/B data. Int. J. Appl. Earth Obs. Geoinf. 34, 188–197.
Phung, H.P., et al., 2020. Monitoring rice growth status in the mekong delta, Vietnam using multitemporal sentinel-1 data. J. Appl. Remote. Sens. 14 (1), 014518.
Rad, A.M., et al., 2019. Developing an automatic phenology-based algorithm for rice detection using Sentinel-2 time-series data. IEEE J. Select. Top. Appl. Earth Observ. 12, 1471–1481.
Richards, F., 1959. A flexible growth function for empirical use. J. Exp. Bot. 10, 290–301.
Sakamoto, T., et al., 2005. A crop phenology detection method using time-series MODIS data. Remote Sens. Environ. 96, 366–374.
Sakamoto, T., et al., 2006. Spatio–temporal distribution of rice phenology and cropping systems in the Mekong Delta with special reference to the seasonal water flow of the Mekong and Bassac rivers. Remote Sens. Environ. 100, 1–16.
Toan, T.L., et al., 1997. Rice crop mapping and monitoring using ERS-1 data based on experiment and modeling results. IEEE Trans. Geosci. Remote Sens. 35, 41–56.
Tornos, L., et al., 2015. Assessment of MODIS spectral indices for determining rice paddy agricultural practices and hydroperiod. ISPRS J. Photogramm. Remote Sens. 101, 110–124.
Veloso, A., et al., 2017. Understanding the temporal behavior of crops using Sentinel-1 and Sentinel-2-like data for agricultural applications. Remote Sens. Environ. 199, 415–426.
Vicente-Guijalba, F., et al., 2014. Crop phenology estimation using a multitemporal model and a Kalman filtering strategy. IEEE Geosci. Remote Sens. Lett. 11, 1081–1085.
Vicente-Guijalba, F., et al., 2015. Dynamical approach for real-time monitoring of agricultural crops. IEEE Trans. Geosci. Remote Sens. 53, 3278–3293.
Wang, S., et al., 2017. Responses of net primary productivity to phenological dynamics in the Tibetan Plateau, China. Agr. For. Meteorol. 232, 235–246.
Yang, Z., et al., 2017. An improved scheme for rice phenology estimation based on time-series multispectral HJ-1A/B and polarimetric RADARSAT-2 data. Remote Sens. Environ. 195, 184–201.
Yang, H., et al., 2018. Field-based rice classification in Wuhua county through integration of multi-temporal Sentinel-1A and Landsat-8 OLI data. Int. J. Appl. Earth Obs. Geoinf. 69, 226–236.
Yuzugullu, O., et al., 2015. Rice growth monitoring by means of X-band co-polar SAR: feature clustering and BBCH scale. IEEE Geosci. Remote Sens. Lett. 12, 1218–1222.
Zhang, X., et al., 2003. Monitoring vegetation phenology using MODIS. Remote Sens. Environ. 84, 471–475.
Zhang, J., et al., 2014. Improved maize cultivated area estimation over a large scale combining MODIS–EVI time series data and crop phenological information. ISPRS J. Photogramm. Remote Sens. 94, 102–113.
Zheng, H., et al., 2016. Detection of rice phenology through time series analysis of ground-based spectral index data. Field Crop Res. 198, 131–139.
Zhong, L., et al., 2016. Automated mapping of soybean and corn using phenology. ISPRS J. Photogramm. Remote Sens. 119, 151–164.